# Advanced Number Theory

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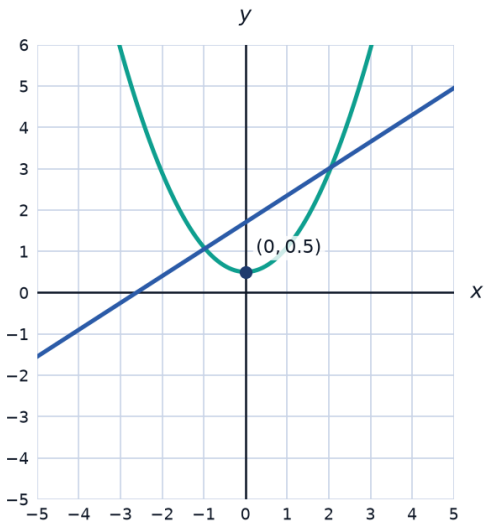
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational] • Target: Divisibility rules and prime factorization**

Find the prime factorization of 1260.

**2** **[Foundational] • Target: Divisibility rules and prime factorization**

Determine whether 143 is prime, and justify your answer.




**3** **[Foundational] • Target: GCF, LCM, and their relationship**

Find the GCF and LCM of 36 and 60.

**4** **[Foundational] • Target: GCF, LCM, and their relationship**

Given that two numbers have GCF 8 and LCM 96, and one of the numbers is 24, find the other number, using the identity GCF × LCM = product of the two numbers.

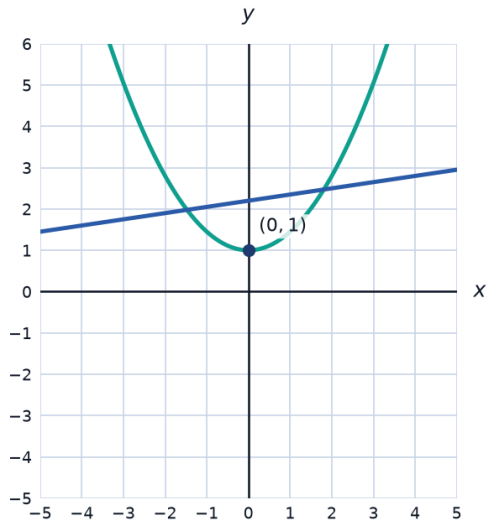
**5** **[Foundational] • Target: Modular arithmetic**

Find the remainder when $47$ is divided by $6$, and express this using modular notation.

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**6** [Foundational] • Target: Modular arithmetic

Find the last digit of $7^{100}$, using the pattern of the last digits of powers of 7 (which cycle every 4: 7, 9, 3, 1).




**7** [Foundational] • Target: Sums and products of integer sets

Find the sum of all integers from 1 to 100, using the formula $S = \frac{n(n+1)}{2}$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Sums and products of integer sets

Find the sum of all even integers from 2 to 200.

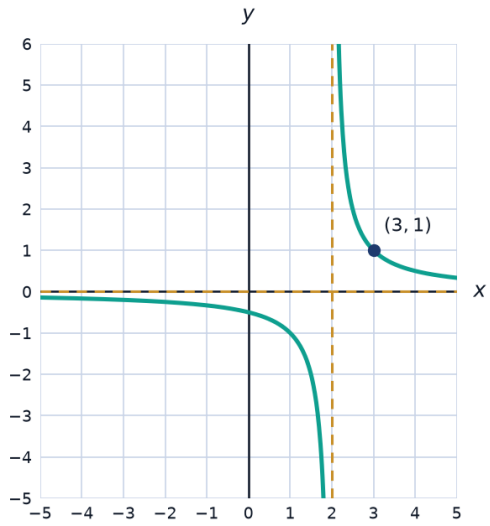
**9** [Intermediate] • Target: Number sets and properties

Classify $\sqrt{16}$ as rational or irrational, and explain your reasoning.

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**10** [Intermediate] • Target: Number sets and properties

Explain why the sum of a rational number and an irrational number is always irrational.




**11** [Intermediate] • Target: Sequences and series

Find the 15th term of the arithmetic sequence with first term 6 and common difference 4.

**12** [Intermediate] • Target: Sequences and series

Find the sum of the first 8 terms of the geometric sequence $2, 6, 18, 54, \ldots$

**13** [Intermediate] • Target: Absolute value and inequalities with integers

Find all integer solutions of $|x - 5| \leq 3$.

**14** [Intermediate] • Target: Combining number theory concepts in a multi-part problem

This question has three parts. (a) Find the prime factorization of 84 and 126. (b) Use the prime factorizations to find the GCF and LCM. (c) Verify your answer using the identity GCF × LCM = product of the two numbers.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

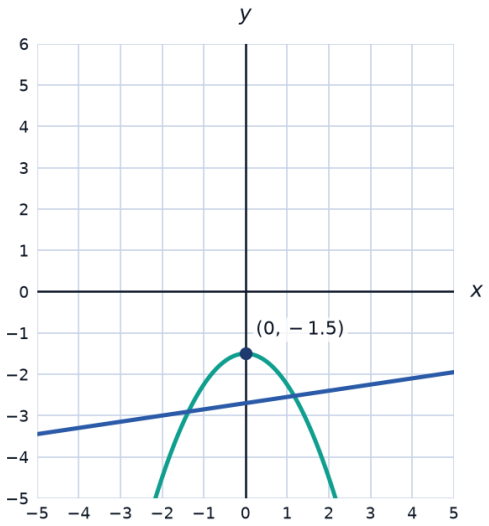
**15** [Challenging] • Target: Perfect squares, perfect cubes, and their properties • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the smallest positive integer $n$ such that $200n$ is a perfect square.

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**16** [Challenging] • Target: Perfect squares, perfect cubes, and their properties • Cross-Lesson: 3.01 Advanced Functions

Determine whether 128 is a perfect cube, and justify your answer using prime factorization.




**17** [Challenging] • Target: Counting divisors of a number • Cross-Lesson: 4.01 Advanced Trigonometry

Find the number of positive divisors of 360, using its prime factorization $360 = 2^3 \times 3^2 \times 5$.

**18** [Challenging] • Target: Working with consecutive integers and integer constraints • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find three consecutive even integers whose sum is 96.

**19** [Challenging] • Target: Working with consecutive integers and integer constraints • Cross-Lesson: 6.01 Advanced Coordinate Geometry

The product of two consecutive positive integers is 156. Find the integers.

**20** [Challenging] • Target: Digit sum and place value problems • Cross-Lesson: 7.01 Advanced Plane Geometry

A two-digit number has digit sum 11. When the digits are reversed, the new number is 27 more than the original. Find the original number.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Divisibility rules and prime factorization

Find the remainder when 128 is divided by 9.

**a.** 1

**b.** 9

**c.** 2

**d.** 3

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**22** [Foundational] • Target: GCF, LCM, and their relationship

Find the 7th term of the geometric sequence $2, 4, 8, 16, \ldots$

**a.** 130

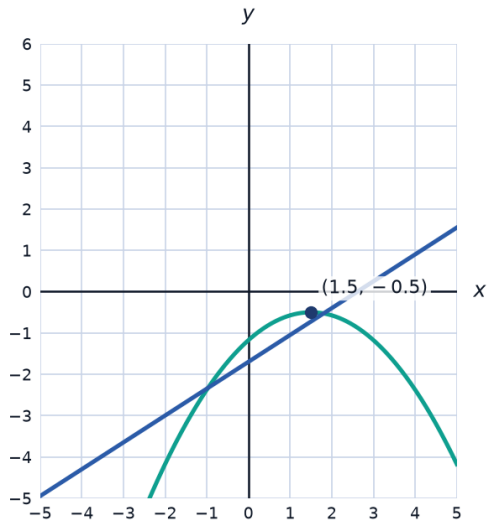
**b.** 125

**c.** 131

**d.** 128

**23** [Foundational] • Target: Modular arithmetic

Let $a$ and $b$ be relatively prime positive integers such that $\frac{a}{b} + \frac{14b}{9a}$ is an integer. What is the sum of all distinct possible values of this integer?




**a.** 8

**b.** 10

**c.** 16

**d.** 24

**24** [Foundational] • Target: Sums and products of integer sets

A positive integer $N$ has the prime factorization $N = 2^a \cdot 3^b \cdot 5^c$, where $a$, $b$, and $c$ are positive integers. If $N$ has exactly 60 positive factors, $2N$ has 72 positive factors, and $3N$ has 80 positive factors, what is the value of $c$?

**a.** 2

**b.** 3

**c.** 4

**d.** 5

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**25** [Foundational] • Target: Number sets and properties

The positive integer $N = 2^a \cdot 3^b \cdot 5^c$, where $a$, $b$, and $c$ are positive integers, has exactly 60 positive divisors. If $\gcd(N, 600) = 120$ and $\gcd(N, 27) = 9$, what is the value of $a + b + c$?

**a.** 6

**b.** 10

**c.** 12

**d.** 13

**26** [Foundational] • Target: Sequences and series

How many ordered pairs of positive integers $(a, b)$ satisfy the equation $\gcd(a, b) + \text{lcm}(a, b) = 105$?

**a.** 20

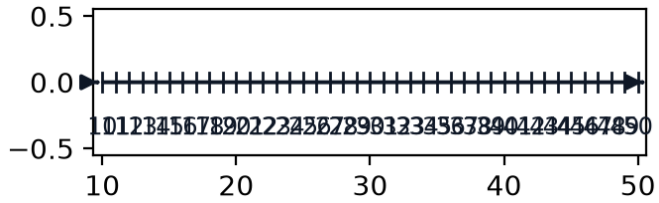
**b.** 24

**c.** 28

**d.** 32

**27** [Foundational] • Target: Absolute value and inequalities with integers

The closed interval $[12, 48]$ is shown on the number line below. How many integers $x$ in this interval satisfy the linear congruence $2x \equiv 1 \pmod 5$?




**a.** 6

**b.** 7

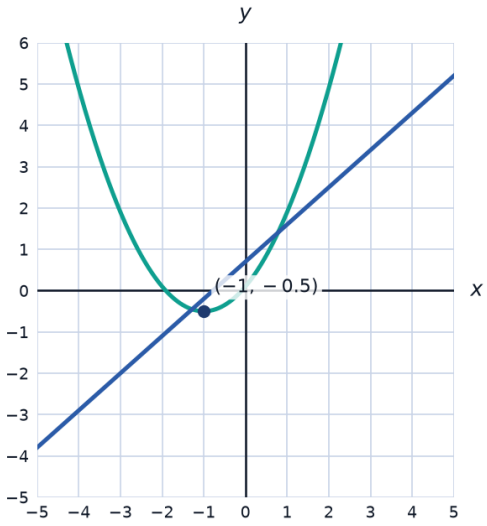
**c.** 8

**d.** 9

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**28** [Foundational] • Target: Combining number theory concepts in a multi-part problem

For a positive integer $k$, the greatest common divisor of $k$ and $24$ is $6$, and the least common multiple of $k$ and $40$ is $600$. What is the value of $k$?




**a.** 75

**b.** 120

**c.** 150

**d.** 300

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Perfect squares, perfect cubes, and their properties

What is the smallest positive integer $n$ such that $126n$ is the square of an integer?

**a.** 7

**b.** 14

**c.** 21

**d.** 42

**30** [Intermediate] • Target: Counting divisors of a number

The polynomial $P(x) = x^4 - 8x^3 + ax^2 + bx + c$ has four real roots that form an arithmetic progression. If the sum of the squares of the four roots is $36$, what is the value of the constant term $c$?

**a.** $-15$

**b.** $-9$

**c.** 9

**d.** 15

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**31** [Intermediate] • Target: Working with consecutive integers and integer constraints

What is the remainder when $4^{103}$ is divided by $11$?

**a.** 9

**b.** 5

**c.** 4

**d.** 3

**32** [Intermediate] • Target: Digit sum and place value problems

Find the greatest common factor (GCF) of 53 and 55.

**a.** 3

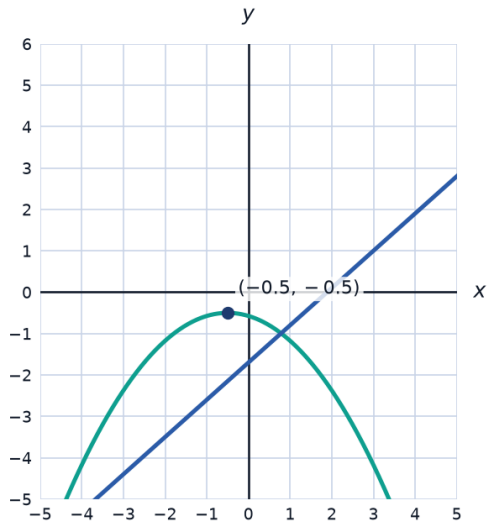
**b.** 2

**c.** 4

**d.** 1

**33** [Intermediate] • Target: A multi-part modular arithmetic and remainder problem

An arithmetic sequence has $a_1 = 3$ and $d = 6$. If the sum of the first $n$ terms is $300$, find $n$.




**a.** 11

**b.** 9

**c.** 5

**d.** 10

**34** [Intermediate] • Target: Divisibility rules and prime factorization

Find the least common multiple (LCM) of 10 and 20.

**a.** 200

**b.** 30

**c.** 20

**d.** 10

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**35** [Intermediate] • Target: GCF, LCM, and their relationship

Find the sum of the first $10$ terms of the arithmetic sequence with $a_1 = 3$ and $d = 1$.

**a.** $-75$

**b.** $150$

**c.** $\frac{165}{2}$

**d.** $75$

**36** [Intermediate] • Target: Modular arithmetic

An arithmetic sequence has $a_2 = 6$ and $a_7 = -4$. Find $a_4$.

**a.** $0$

**b.** $10$

**c.** $2$

**d.** $-2$

**37** [Intermediate] • Target: Sums and products of integer sets

What is the common difference of the arithmetic sequence $0, -2, -4, -6, \ldots$?

**a.** $0$

**b.** $-2$

**c.** $-4$

**d.** $-3$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Number sets and properties • Cross-Lesson: 5.01 Pre-Calculus Concepts

An arithmetic sequence has first term $a_1 = -7$ and common difference $d = -1$. Find $a_6$.

**a.** $-2$

**b.** $-12$

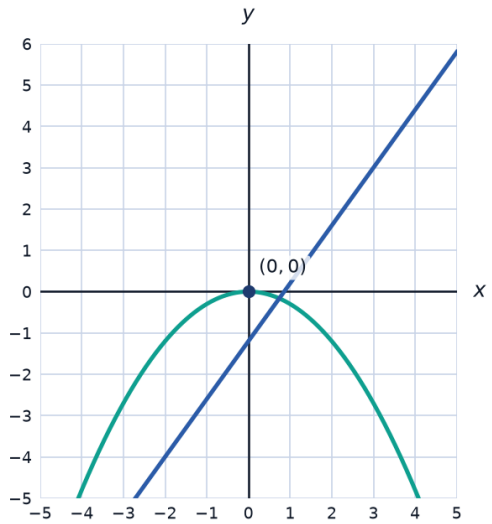
**c.** $12$

**d.** $-13$

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**39** [Challenging] • Target: Sequences and series • Cross-Lesson: 6.01 Advanced Coordinate Geometry

An infinite geometric sequence $a_1, a_2, a_3, \ldots$ of positive real numbers has common ratio $r < 1$ and sum $S$. The sub-sequence of terms whose indices are of the form $3k - 1$ for positive integers $k$ (that is, $a_2, a_5, a_8, \ldots$) has sum $\sum_{}^{\infty} a_{3k-1} = \frac{3}{13}S$. Furthermore, the infinite series formed by the squares of the terms has sum $\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S$. What is the value of $\sum_{n=1}^{\infty} na_n$?




**a.** $\frac{27}{2}$

**b.** $\frac{81}{4}$

**c.** $\frac{9}{2}$

**d.** $\frac{27}{4}$

**40** [Challenging] • Target: Absolute value and inequalities with integers • Cross-Lesson: 7.01 Advanced Plane Geometry

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference $d$. The terms $a_1$, $a_3$, and $a_4$ form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the $n$-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?

**a.** $\frac{5}{4}$

**b.** $\frac{81}{64}$

**c.** $\frac{9}{8}$

**d.** $\frac{5}{8}$

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**41** [Challenging] • Target: Combining number theory concepts in a multi-part problem • Cross-Lesson: 8.01 Statistics, Counting & Probability

Prove $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ using the formula for arithmetic sums.

**a.** Cannot be proved simply

**b.** By FTA

**c.** By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation

**d.** Use telescoping

**42** [Challenging] • Target: Perfect squares, perfect cubes, and their properties • Cross-Lesson: 2.01 Advanced Algebra & Systems

Use generating functions to find the number of ways to write $n$ as an ordered sum of positive integers.

**a.** $n^2$

**b.** $2^{n-1}$

**c.** $n!$

**d.** $2^n$

**43** [Challenging] • Target: Counting divisors of a number • Cross-Lesson: 3.01 Advanced Functions

A storage bin contains red, blue, and green blocks. The ratio of red blocks to blue blocks is $3:5$. After 12 green blocks are added to the bin, with no other blocks added or removed, the total number of blocks in the bin is 60, and the probability of randomly selecting a green block from the bin is $\frac{1}{3}$. How many red blocks were in the bin originally?

**a.** 12

**b.** 15

**c.** 18

**d.** 25

**44** [Challenging] • Target: Working with consecutive integers and integer constraints • Cross-Lesson: 4.01 Advanced Trigonometry

The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music?

**a.** $\frac{7}{50}$

**b.** $\frac{7}{20}$

**c.** $\frac{7}{13}$

**d.** $\frac{13}{20}$

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**45** [Challenging] • Target: Digit sum and place value problems • Cross-Lesson: 5.01 Pre-Calculus Concepts

What is the sum of all real values of $k$ for which the equation $|x^2 - 4| = 2x + k$ has exactly three distinct real solutions?

**a.** 4

**b.** 5

**c.** 7

**d.** 9

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